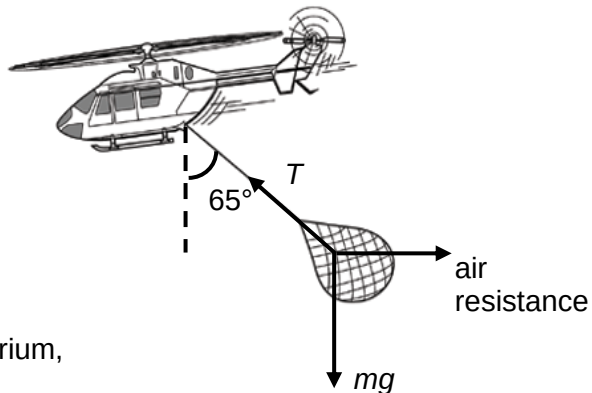


2(a)(i)	Applying Newton's second law, $T - (4 \times 10^3)(9.81) = (4 \times 10^3)(0.32)$ $T = 4.05 \times 10^4 \text{ N}$	[1] for correct substitution [1] for answer
2(a)(ii)1.	mass per unit time = ρ (vol per unit time) $= \rho(Av)$ $= (1.3) \left[\pi (10)^2 \right] v$ $= 408v$ $= 410v$	[1] for correct substitution [1] for correct intermediate value
2(a)(ii)2.	Applying Newton's second law, $F_{net} = \frac{dp}{dt} = \left(\frac{dm}{dt} \right) v$ $9.1 \times 10^4 = (410v)v$ $v = 14.9 \text{ m s}^{-1}$	[1] for correct substitution [1] for correct answer

2(b)



For vertical equilibrium,

$$T \cos 65^\circ = mg = (4 \times 10^3)(9.81) \quad \text{--- (1)}$$

Horizontally,

$$T \sin 65^\circ - 6 \times 10^3 = ma = (4 \times 10^3)a$$

$$T \sin 65^\circ = (4 \times 10^3)a + 6 \times 10^3 \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} : \tan 65^\circ = \frac{ma + 6 \times 10^3}{mg}$$

$$a = 19.5 \text{ m s}^{-2} = 20 \text{ m s}^{-2}$$

[1] for correct substitution

[1] for correct answer

3(a)(i)

The moment of a force about a pivot is the product of the force

[1]